

# **SMART INDIA HACKATHON 2026 EVALUATION REPORT**

**AI POWERED SEARCH & RESCUE  
DRONE**

***Problem  
Statement  
ID: SIH26177  
/ Category:  
Hardware /  
Theme:  
Robotics and  
Drones***

| TEAM NAME                | PROBLEM STATEMENT | DOMAIN THEME      |
|--------------------------|-------------------|-------------------|
| DRONE DYNAMICS HUB (DDH) | SIH26177          | ROBOTICS & DRONES |

## 1. Executive Summary

Drones During natural disasters and emergency search-and-rescue (SAR) operations, traditional manual search methods are often constrained by hazardous terrain, low visibility, and critical time delays. Drone Dynamics Hub (DDH) proposes a deployable, AI-powered autonomous UAV solution engineered to dramatically reduce victim discovery time, enhance responder safety, and deliver live situational intelligence. □

***"Our drone doesn't just find people — it tells rescuers WHO, WHERE, WHAT RISK, and WHERE TO RESPOND FIRST."***

## 2. Proposed Solution & Core Capabilities

**The proposed platform extends beyond raw video streaming by converting live aerial feeds into actionable, prioritized rescue intelligence:**

- ❖ **Dual-Mode Navigation (GPS + SLAM):** Real-time location tracking and path planning using GPS; automatically transitions to SLAM-based navigation in GPS-denied or damaged environments.
- ❖ **Real-Time AI Detection (YOLO + OpenCV):** Deep learning models run on-edge to detect victims and environmental hazards (fire, smoke, debris, damaged structures).
- ❖ **Geo-Tagged Mapping:** Automatically logs exact coordinates for all detected victims and risks, building an interactive disaster map.
- ❖ **Rescue Information Dashboard:** Displays live priority-ranked alerts, status updates, and clear access routes to ground response teams.
- ❖ **Multi-UAV Fleet Management:** Scalable architecture supporting coordinated multi-drone operations for large disaster areas.
- ❖ **Adaptive Search Strategy:** Combines broad-area long-range scanning with close-range verification routines.
- ❖ **Offline-First Resilience:** Local storage of detection and mapping logs ensures uninterrupted operation during communications blackout.

## **IMPORTANT INNOVATION AND UNIQUE IDEA**

**1. Cellular 4G/5G — Cloud Connectivity Available:** The cellular network can help locate an active mobile device associated with a victim using information such as Cell ID , timing, and signal measurements. The information can be sent through the network to a cloud/server and displayed on the ground station dashboard.

**2. Cellular 4G/5G — Cloud Connectivity Unavailable:** When the cloud is unavailable, edge computing can process cellular measurements locally on the UAV. The estimated location, timestamp, and confidence can be stored on the UAV and transferred to the ground station when it returns.

**3. Multiple UAV Coordination — Cloud Available:** Multiple UAVs can send their telemetry and victim detections to a central cloud service and shared database. A single dashboard can then display UAV positions, battery status, mission status, and detected victims.

**4. Multiple UAV Coordination — Cloud Unavailable:** Without cloud connectivity, UAVs can communicate through local networks, Wi-Fi/mesh, LoRa, private LTE/5G, or radio telemetry. Each UAV can scan its assigned area, process data locally, and exchange mission information with other UAVs and the ground station.

**5. UWB / mmWave Radar — RF Signal Transmission** Radar transmits RF signals toward the target and receives the reflected signal for processing. mmWave/FMCW radar can estimate distance and movement using frequency and Doppler information, while UWB uses wide-band signals to detect presence, range, and movement.

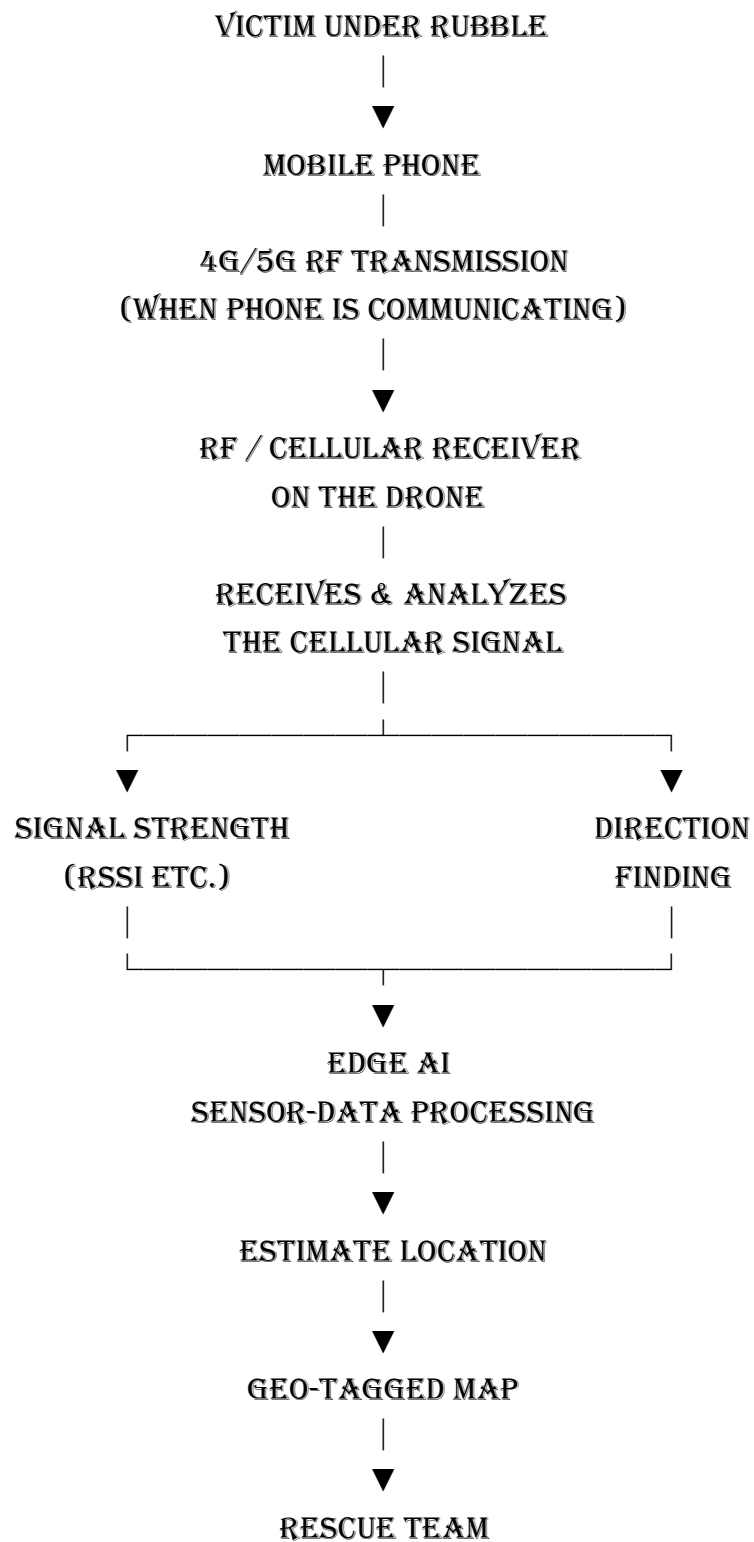
**6. No Cloud — Storing Victim Location on SD Card** The UAV can combine radar/sensor detection with GNSS coordinates to store a geotagged victim record on an SD card. When the UAV returns, the data can be transferred to the ground station and displayed on an offline map.

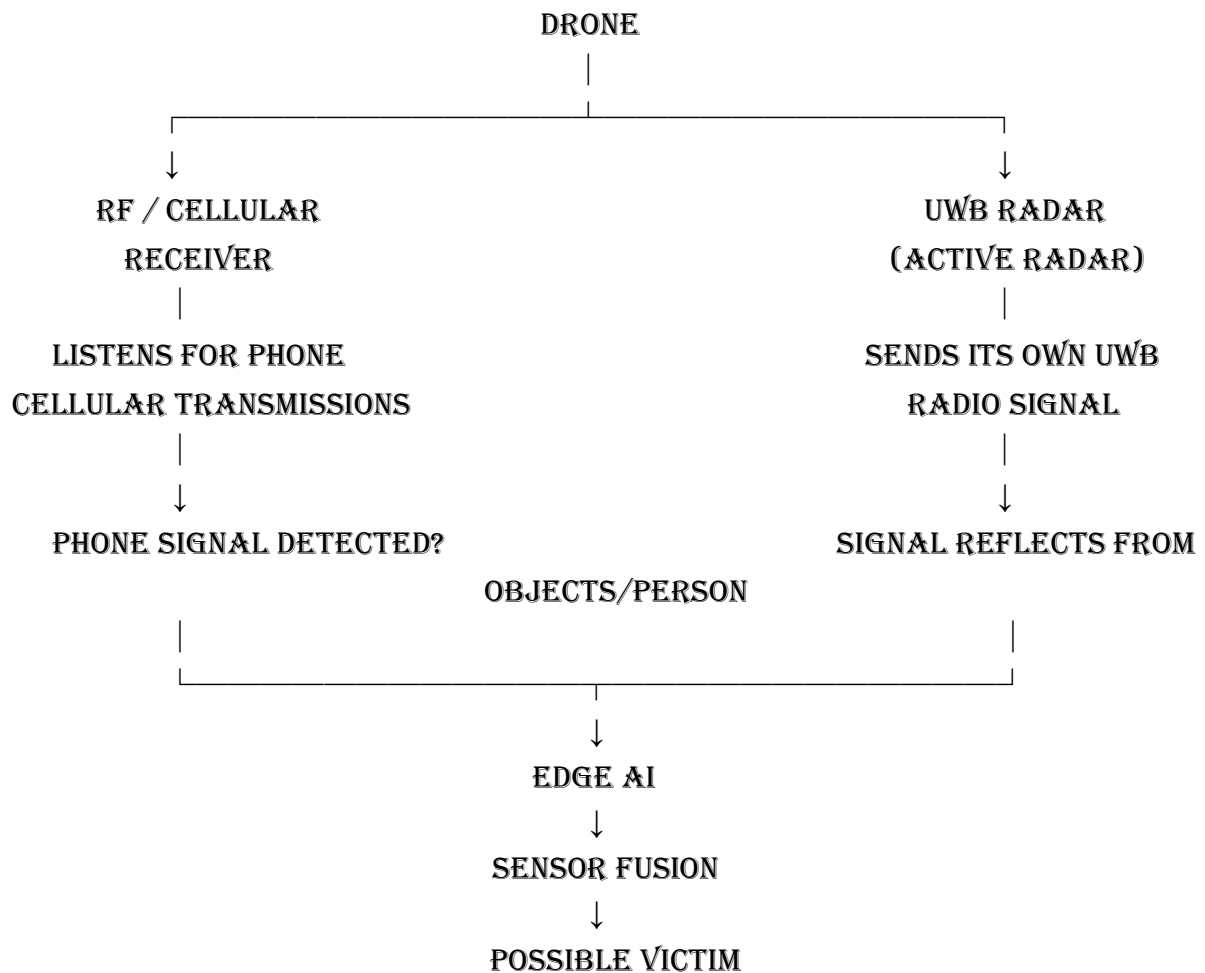
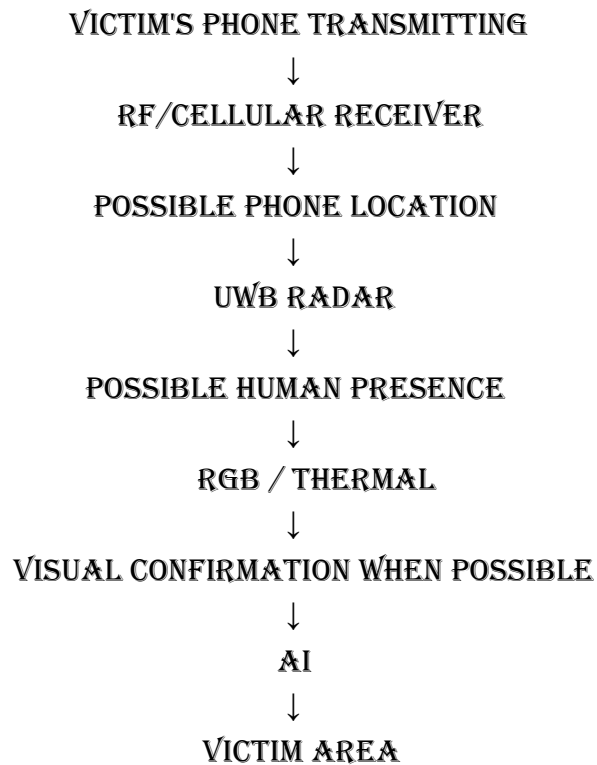
**Overall Architecture:** The system can be divided into Cloud, Ground, and UAV Edge layers. If the internet fails, the system switches from cloud synchronization to local communication and offline processing, allowing the mission to continue.

**Sensor Fusion:** The system can combine cellular indication, UWB/mmWave radar, camera and thermal sensing to improve victim-detection confidence. The fused result can be linked with GPS coordinates and stored locally or displayed on the dashboard.

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### Proposed solutions in flowchart





### 3. Technical Approach & Architecture System

*Stack Breakdown*

| LAYER                | COMPONENTS & TECHNOLOGIES                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------|
| Hardware & Avionics  | Flight Controller, IMU, GPS, Telemetry Radio, Digital Camera, Directional Acoustic Microphone Array |
| Software & OS        | Ubuntu Linux, Python Runtime, MAVLinkProtocol for Flight Control Integration                        |
| AI & Computer Vision | PyTorch, OpenCV, YOLO Architecture (OnboardReal-time Edge Detection)                                |
| Future Sensors       | Thermal Imaging (Darkness/Smoke Penetration), LiDAR, Radar Sensor Fusion                            |

❖ **END-TO-END MISSION WORKFLOW**

- ① Disaster Area Survey: System initialized and mission parameter targets defined.
- ② Autonomous Deployment: Launch with GPS navigation or SLAM pathing.
- ③ Sensor Data Collection: Continuous HD video stream and acoustic pulse detection.
- ④ AI Detection & Classification: Onboard YOLO detects trapped individuals and active hazards.
- ⑤ Geo-Tagging & Mapping: Coordinates stamped onto localized maps.
- ⑥ Priority Ranking: Automated algorithm ranks critical cases based on severity and risk.
- ⑦ Dashboard Broadcast: Ground teams receive live priority alerts and safe access routes.
- ⑧ Responder Dispatch: Targeted rescue execution minimizing team exposure to hazards.

## 4. Feasibility, Key Challenges & Mitigations

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| <u>CHALLENGE<br/>CATEGORY</u> | <u>OPERATIONAL RISK</u>                            | <u>MITIGATION<br/>STRATEGY</u>                               |
|-------------------------------|----------------------------------------------------|--------------------------------------------------------------|
| Autonomous Navigation         | GPS blockage in urban canyons or inside structures | SLAM-based localization and camera-driven obstacle avoidance |
| Low Visibility                | Thick smoke, dust, or night-time operations        | Multi-sensor fusion incorporating thermal imaging and LiDAR  |
| Common Failure                | Cellular network or RF telemetry disruption        | Edge AI processing with local offline data persistence       |
| Battery Limits                | Restricted flight duration on active missions      | Edge AI processing with local offline data persistence       |

## 5. Development Roadmap

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- **Phase 1: Prototype Development** — Integration of camera-based YOLO detection, OpenCV pipeline, and basic MAVLink telemetry.
- **Phase 2: Navigation & Sensor Testing** — Implementation of SLAM navigation, directional acoustic sensors, and RF localization.
- **Phase 3: Field Trials & Mock Drills** — Collaborative trial testing with NDRF and SDRF teams in simulated disaster zones.
- **Phase 4: Pilot Deployment** — Full real-world rollout in active disaster management operations.

## 6. Impact, Business Model & SDG Alignment

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- **Primary Users**: NDRF, SDRF, Fire & Emergency Services, Police, Disaster Management Authorities, Civil Defence, and emergency NGOs.
- **Financial & Funding Model**: Government disaster management schemes, grants (DRDO, ISRO), CSR initiatives, Public-Private Partnerships (PPP), and long-term service contracts.
- **UN Sustainable Development Goals (SDGs)**:

*SDG 3: Good Health & Well-Being*

*SDG 11: Sustainable Cities & Communities*

*SDG 13: Climate Action*

## 7. Future Research & References

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Future Directions: Thermal vision for zero-visibility scenarios, advanced multi-sensor fusion (RGB+Thermal+LiDAR+IMU), and edge model quantization for ultra-fast onboard inference.

### *References*

1. PX4 Autopilot Documentation — <https://docs.px4.io/>
2. ROS 2 Documentation — <https://docs.ros.org/>
3. Ultralytics YOLO Documentation — <https://docs.ultralytics.com/>
4. OpenCV Documentation — <https://docs.opencv.org/5.0/>
5. Gazebo Simulation Docs — <https://gazebo.org/>
6. Smart India Hackathon Portal — <https://sih.gov.in/>
7. NDMA Disaster Management Guidelines — <https://ndma.gov.in/ndma-guidelines>
8. IEEE Papers on UAV Search & Rescue — <https://ieeexplore.ieee.org/>

## 8. Key Proposal Highlights & Executive Summary Points

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- 1) **Autonomous UAV Search & Rescue System:** Designed to accelerate victim discovery time and protect emergency responders in complex disaster zones.
- 2) **Real-time AI Detection (YOLO + OpenCV):** Edge deep-learning framework detects humans, fires, smoke, debris, and structural damage directly from the drone feed.
- 3) **Geo-Tagged Hazard Mapping:** Automatically logs precise GPS coordinates for detected victims and environmental hazards to construct live situational maps.
- 4) **Priority-Ranked Dashboard:** Computes dynamic risk algorithms to present rescue commanders with prioritized victim dispatch orders and safe access routes.
- 5) **GPS & SLAM Dual-Navigation:** Seamlessly transitions from GPS pathing to indoor/obstacle SLAM navigation in GPS-denied environments.
- 6) **Directional Acoustic Localisation:** Employs specialized microphone arrays to detect SOS acoustic pulses, screams, and whistles.
- 7) **Offline-First Infrastructure:** Guarantees operational reliability through local data processing and persistence during total telemetry or network blackouts.
- 8) **End-to-End Operational Feasibility:** Supported by open-source robotics tools (PX4, ROS 2) and targettested for NDRF/SDRF emergency team integration.
- 9) **Sustainable Strategic Alignment:** Directly aligned with UN Sustainable Development Goals 3, 11, and 13 while backed by CSR, PPP, and government grant models.

# SEARCH AND RESCUE PROTOTYPE

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## 1) DRONE PROTOTYPE



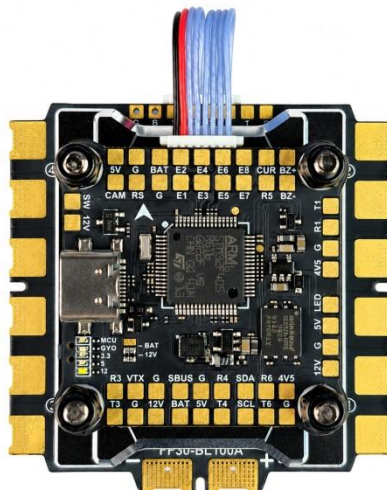
# COMPONENTS

For drone

1. Dji o4 pro video system



2. Aacoda rc f405 flight controller stack



3) 2300kv bldc motors



4.) M10 gps



5. Radiomaster rp1 elrs receiver



6. Pocket elrs transmitter



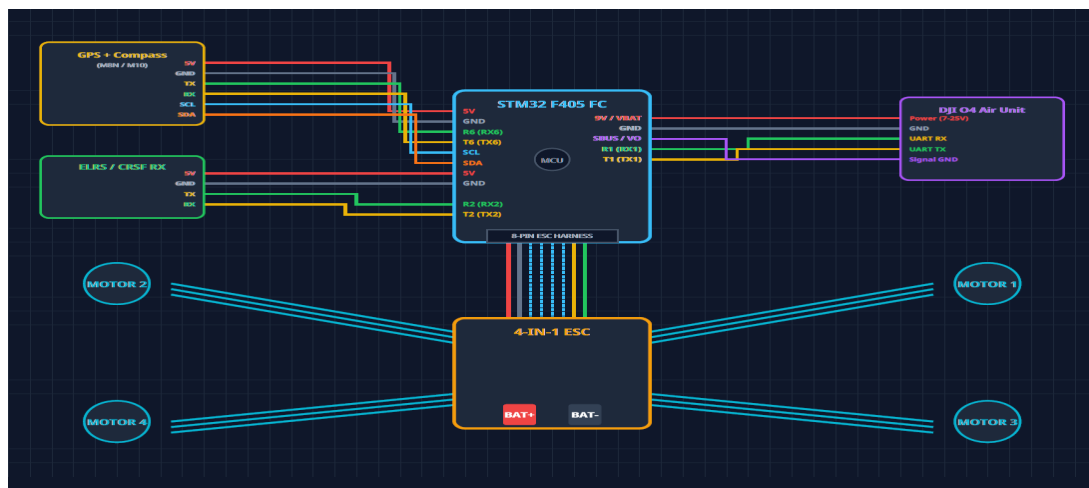
### 7. Dji goggles n3



### 8. Speedybee 5" frame

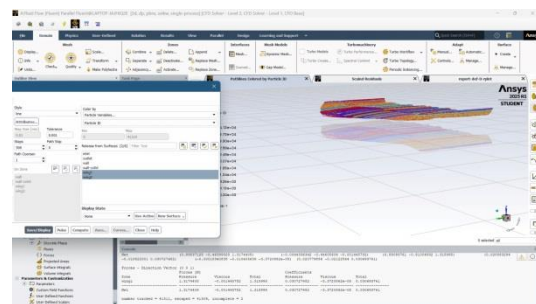
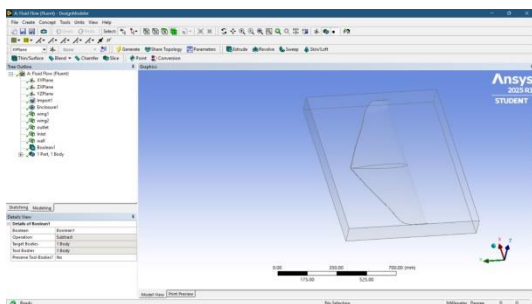
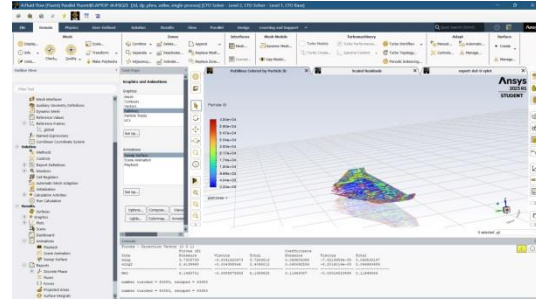
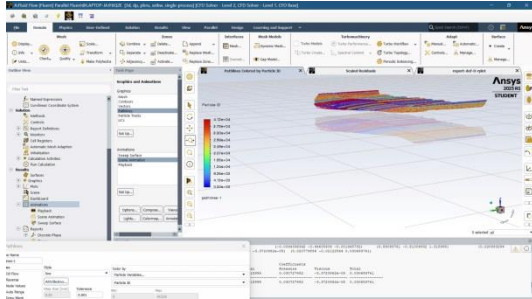


## CIRCUIT DIAGRAM

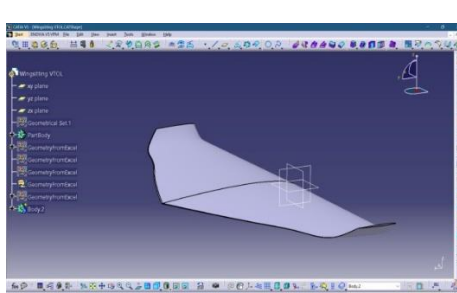
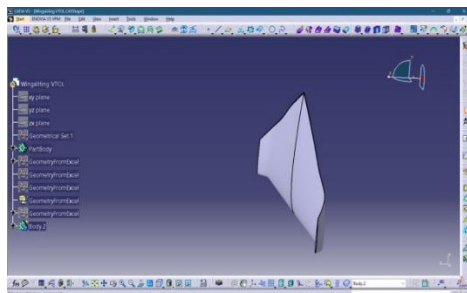
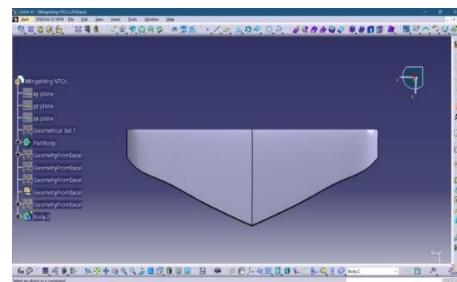
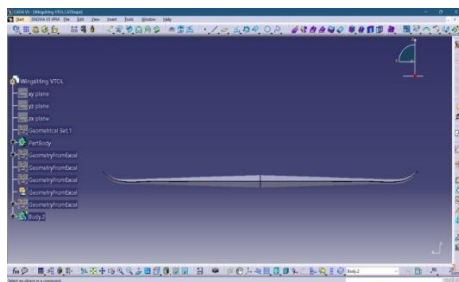


## 2) FT SPEAR

### 2.1) PROTOTYPE ANALYSIS



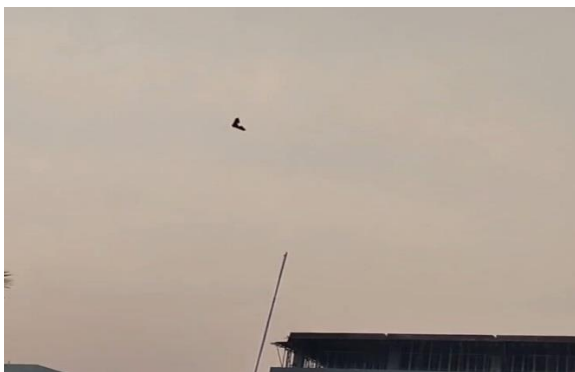
### 2.2) CAD MODELING



# PROTOTYPE

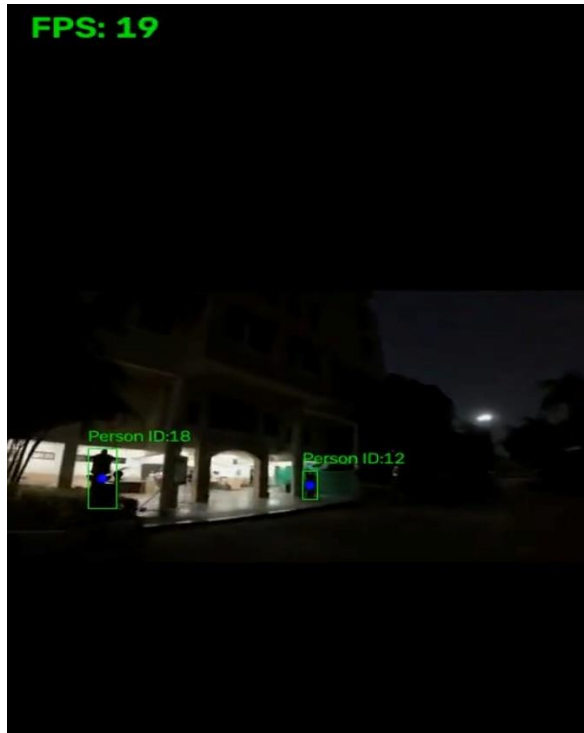


## FT SPEAR AND DRONE FLIGHT





# YOLO DETECTION FOR VICTIM AND HAZARD DETECTION



# GPS NAVIGATION



*“Return to Home Autonomously through use of GPS Navigation from upto 2km”*



*"We are trying to integrate some innovative concepts or Ideas in it, search and Rescue that can help in the future. Working on an advanced Search & Rescue solution by integrating multiple sensors, cellular technologies, and multi-drone coordination through a common dashboard, enabling our prototype drone and FT Spear to communicate and share data seamlessly"*